Brett M. Hixon

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SUMMARY

High-achieving computer science senior (3.98 GPA) and Co-Founder of Fafnir Aerospace with proven industry experience in scalable software architecture and autonomous drone systems. Merges a broad academic foundation—spanning AI, cybersecurity, and Japanese—with core programming (C, C++, C#, Java, Python) to engineer robust systems, from enterprise-level infrastructure to AI-powered mesh networks.

EDUCATION

Rose-Hulman Institute of Technology, Terre Haute, IN *GPA: 3.98/4.0*

Bachelor of Science, Computer Science — May 2027

Minors in Cybersecurity, Artificial Intelligence, and Japanese

Related Courses: Software Requirements Engineering, Software Design, Machine Learning, Computer Networks, Parallel Computing, Operating Systems, Database Systems, Image Recognition, Foundations of Cybersecurity, Cryptography, Malware Analysis

SKILLS

Software: Java, C, C#, C++, ARM Assembly, JavaScript, Verilog, Python, Lua, RISC-V Assembly, Scheme

Frameworks & Technologies: .NET (ASP.NET Core), Entra ID, ROS 2, Docker, Entity Framework Core, SQL, React, React Native, HTML, CSS, Vue 3, Tailwind, PyTorch, pandas, NumPy, scikit-learn

Systems: Windows, macOS, Linux

Language: Limited Working Proficiency in Japanese and Elementary Spanish

EXPERIENCE

Steel Dynamics, Inc. — Software Development Intern *June 2026 - August 2026*

- Rebuilt a legacy PowerBuilder general ledger reporting tool as a modern web application in Vue.js secured by Entra ID as the frontend engineer and integrated with an ASP.NET Core backend connected to SQL Server-backed Microsoft Dynamics GP financial data
  - Reduced new user onboarding to the tool from around 1 hour to just a few minutes
- Designed and deployed a .NET Windows Service to automate SQL Server Extended Events change-approval ticket linkage for quarterly audits as part of a two-person team
  - Eliminated about 8-16 hours total of manual reconciliation work every fiscal quarter
- Partnered with full-time engineers, IT personnel, and financial stakeholders to gather requirements and validate program output

Fafnir Aerospace — Co-Founder *April 2026 – Present*

- Co-founded a venture engineering a proprietary UV light communication module to enable autonomous drone mesh networking in RF-denied and military jamming environments
- Architecting a decentralized multi-drone navigation stack in ROS 2, integrating OpenVINS and Swarm-SLAM for spatial mapping and pose estimation while generating point cloud cost maps for dynamic routing and collision avoidance
- Developing a real-time computer vision pipeline that fuses YOLO object detection with SLAM geometric coordinates to autonomously identify, classify, and map targets

Rose-Hulman Ventures — Software Engineer Intern *January 2025 - May 2025*

- Researched and practiced multiple web and mobile app frameworks such as React/React Native, ASP.NET Core, and Entity Framework Core
- Created internal documentation for future engineer interns to reuse when getting started
- Gained experience in full-stack app development and integrated databases in a collaborative environment
- Refactored data retrieval methods for an internal web-based inventory management system built in React

PROJECTS

Recognizing Human Activity from Radar *July - August 2026*

- Developed a deep learning pipeline to classify human activities from FMCW radar data
- Implemented a denoising algorithm for micro-Doppler spectrograms in the preprocessing stage to reduce White Gaussian thermal noise
- Engineered and evaluated a baseline PyTorch CNN model achieving 88% classification accuracy across 11 activity classes
- Collaborated to dissect and integrate a novel CRCNN architecture to further enhance spectrogram classification performance

RAFT Consensus over Drone Mesh Network *May 2026*

- Implemented RAFT consensus protocol with leader election and log replication across containerized nodes in Python and Docker
- Built multithreaded infrastructure to manage leader/election timeouts and follower log catch-up
- Developed test suites validating consensus correctness under simulated drone mesh network conditions